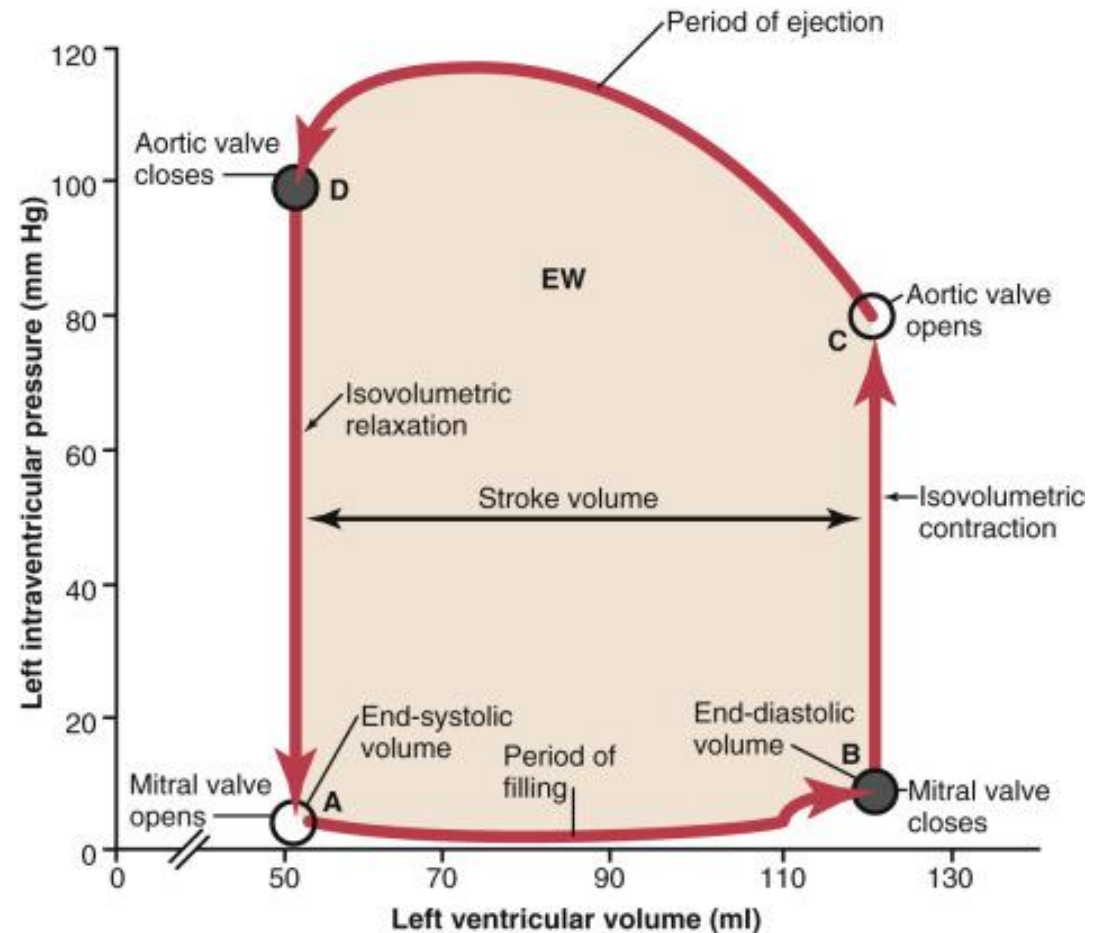
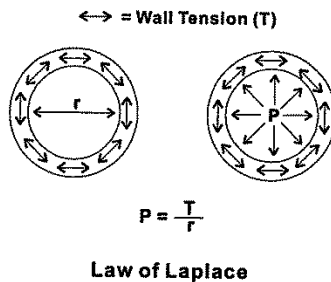


Session Objectives

1. Graph the relationship between cardiac pressures and volumes using the Law of Laplace
2. Recognize concepts of cardiac output and stroke volume in a cardiac cycle
3. Discriminate how afterload, preload (Frank-Starling's Law), and contractility regulate stroke volume and cardiac output in cardiac cycle
4. Apply Fick principle in the measurement of cardiac output
5. Extrapolate the phases, loops, laws, and cardiac output regulation in representative cardiovascular pathologies

Pressure-Volume Curve and Laplace Law

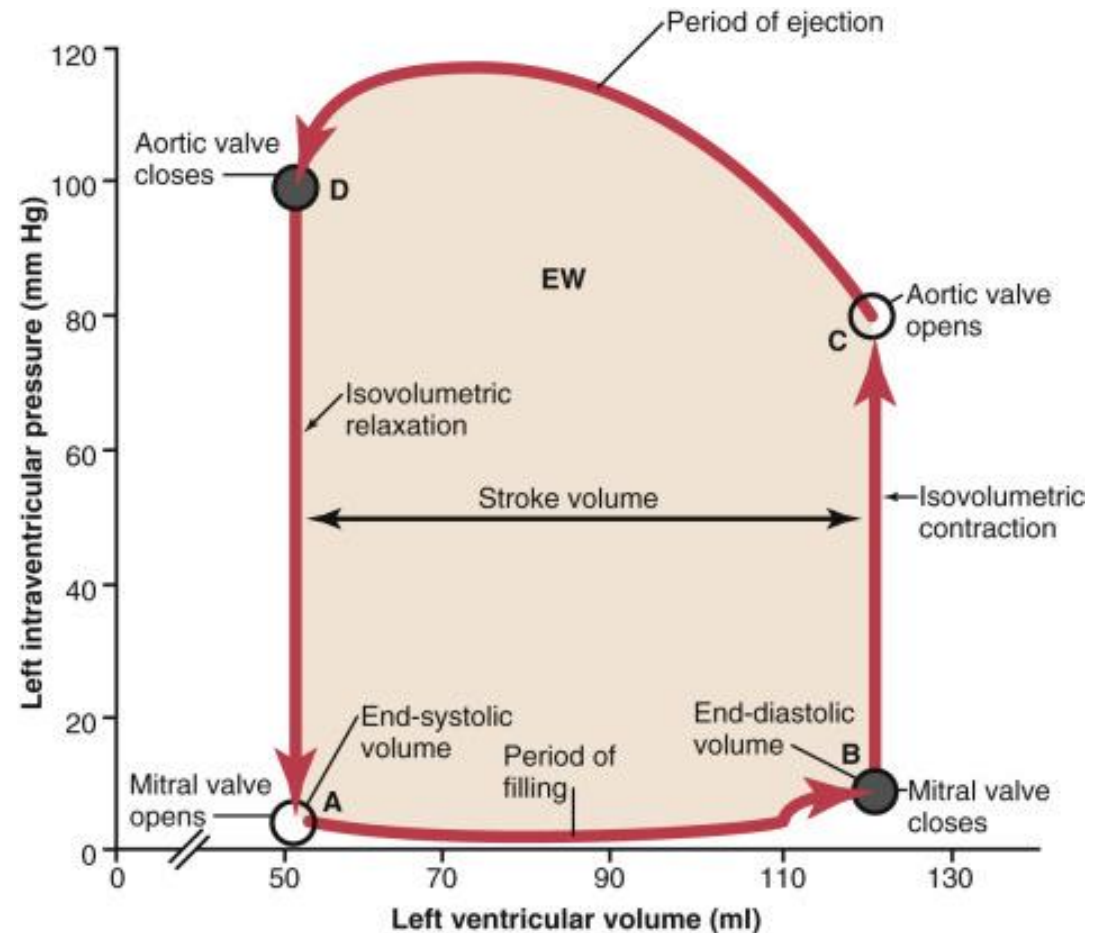
- LAW OF LAPLACE** relates **ventricular pressure (P)** to the **tension (T)** in the ventricular wall divided by the **ventricular radius (r)**.



Pressure-Volume Curve and Laplace Law (contd...)

- The **end diastolic volume (EDV)** is referred to as **preload**.
- The **afterload** is proportional to **arterial pressure**

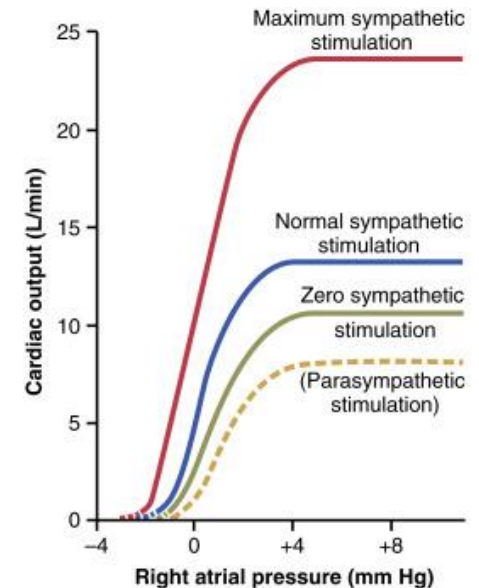
$$\sigma \propto \frac{P \times r}{h}$$



Regulation of HR and SV and CO

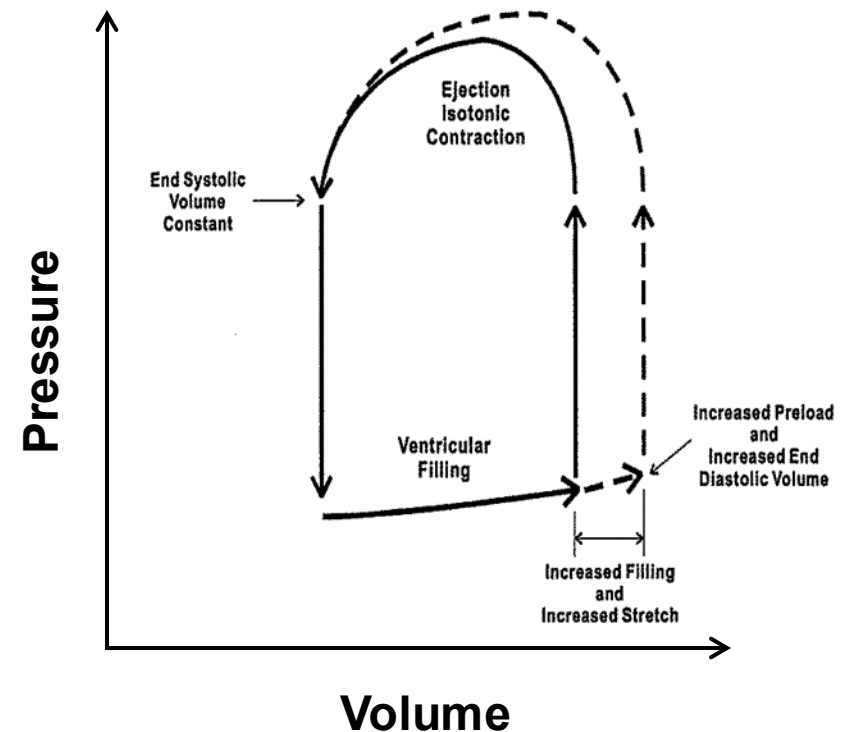
HR can be changed by sympathetic and parasympathetic stimulation.

- **Increase in HR decreases SV** (decreases the amount of time the heart spends in diastole) and thus **the CO is maintained**.
 - However, as HR increases to about **180-200 bpm, CO increases**. This is because the increase in HR more than compensates for the decrease in SV.
 - **Beyond** this range both stroke volume and **CO decrease**.
- Goal in changing HR and/or SV is to aid in **maintaining the mean arterial pressure (MAP)**. **Constriction of arterioles and veins** also aid in maintaining BP.
 - **MAP = CO x TPR (Total peripheral resistance)**



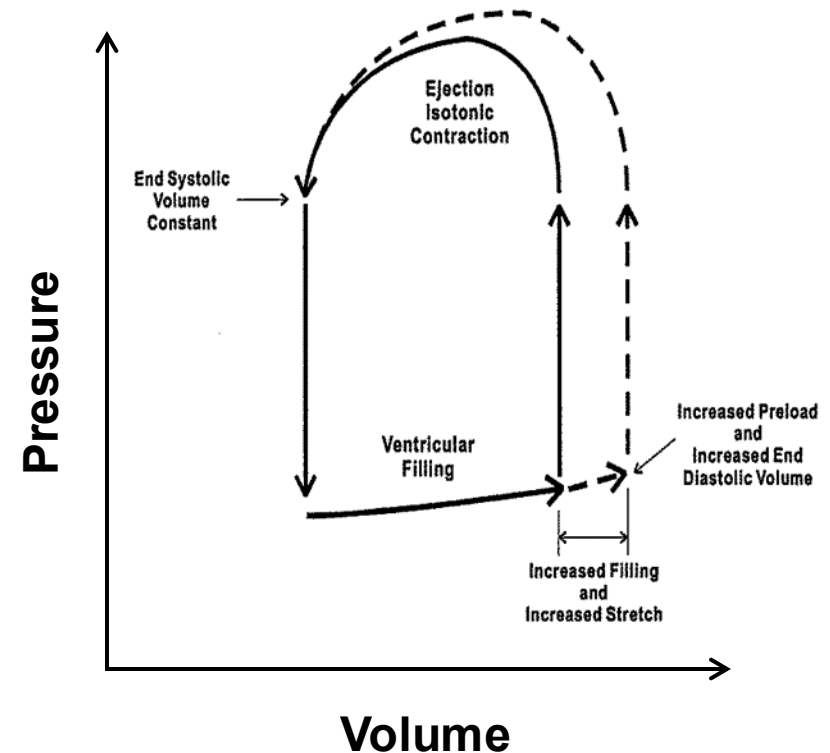
Frank-Starling Law – Effect of Preload (contd)

- The **FRANK-STARLING LAW** (Starling's Law of the Heart) states that SV increases as ventricular EDV increases.
- The greater the passive stretch or preload of the ventricle, the greater the force of ventricular contraction, and the greater the SV.



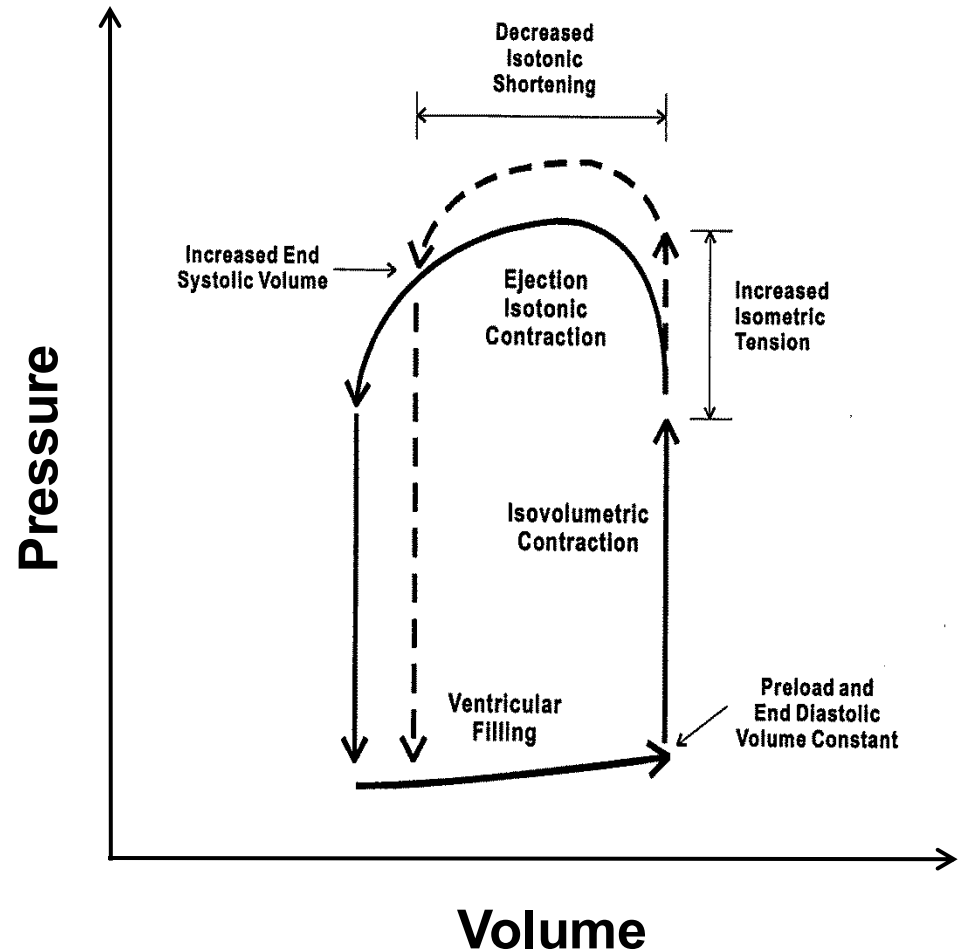
Frank-Starling Law – Effect of Preload (contd)

- **Venous vasoconstriction** contributes to increased **venous return (VR)** and increased **ventricular stretch** → enough force to eject the "extra" returned blood. ESV is not changed.
- This normally **prevents pooling** of blood in pulmonary circulation and systemic circulation (e.g., postural changes).



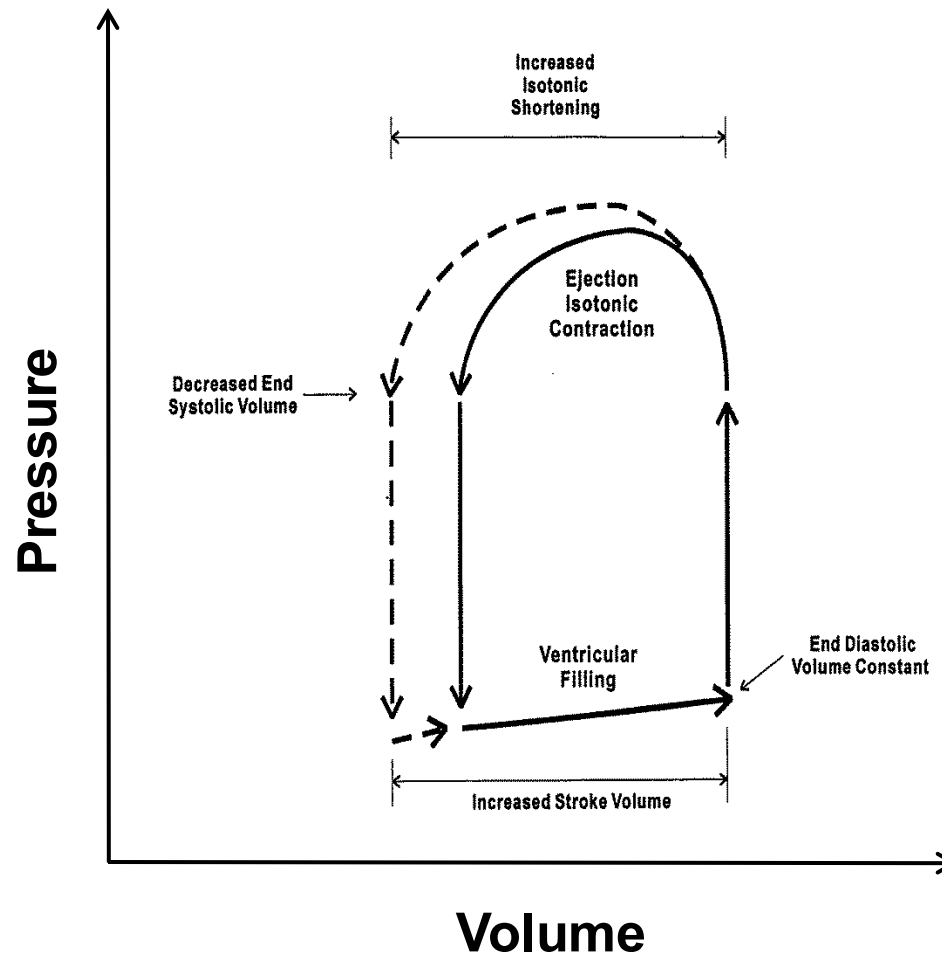
Effect of Afterload

- Afterload is the arterial pressure (AO or PA) that the heart must overcome to circulate blood.
- The greater the afterload, the smaller the SV. The decrease in SV causes an increase in ESV (e.g. more blood remains in the heart) culminating in **heart failure (HF).**
- Hypertension can increase **afterload** and decrease SV and thus CO.



Changes in Contractility – Effect of Norepinephrine

Contractility is an **intrinsic** measure of cardiac performance augmented by factors like Norepinephrine



Sample question

According to Starling's Law of the heart, cardiac output is directly related to:

- a) the size of the ventricles
- b) the heart rate
- c) amount of blood returning to the heart
- d) end-systolic volume
- e) cardiac reserve

Cardiac Output variables

- Factors that change **stroke volume** include:
 - **P**reload (Frank-Starling Law)
 - Venous vasodilators ↓ preload
 - **A**fterload (~ MAP)
 - Arterial vasodilators ↓ afterload
 - ACE inhibitors and ARBs ↓ preload and afterload
 - Chronic hypertension leads to LV hypertrophy
 - **C**ontractility (intrinsic)
 - beta1-blockade, HFrEF, acidosis, ↓PO₂/↑PCO₂, Non-dihydropyridine Ca⁺⁺ channel blockers ↓ contractility
- **Myocardial O₂ demand** is directly proportional to:
 - **C**ontractility
 - **A**fterload
 - Heart **R**ate
 - **D**iameter of ventricle (wall tension)
- **Ejection fraction**
 - ↓ HFrEF
 - Preserved/Normal in HFpEF

Case study

- A 73-year old man visits ER with complaints of exertional dyspnea, syncope, chest pain. Poorly managed hypercholesterolemia with 11-year history of hypertension.
 - BP: 101/75.
 - Lungs clear. Systolic murmur loudest over the aorta; peaks at mid systole.
 - ECG: LAD, ST normal.
 - CXR: large LV, calcification of aortic valve.
 - Echo: Greatly increased velocity of flow during the systolic portion of the cardiac cycle. Left heart enlarged and LVH.
 - Cardiac enzymes normal.
-
- a) Anti-hypertensive therapy
 - b) Coronary artery bypass grafting
 - c) Percutaneous coronary intervention
 - d) Aortic valve repair/replacement
 - e) Sublingual nitroglycerin

Summary Review

- **Law of Laplace**
- **Ventricular Pressure-Volume Curve, DCM**
- **Cardiac Output**
- **Regulation of Stroke Volume**
 - **Preload (Frank-Starling Law)**
 - **Afterload**
 - **Contractility and Calcium**
- **Clinical Conditions – MI, valvular pathology**